# Part I

# Circular Oversampling Methods

# Chapter 1

# Proposed Circular Oversampling Methods

This chapter presents the three proposed circular oversampling algorithms in full mathematical detail. We begin with the common preprocessing pipeline shared by all methods, then present each algorithm with its formulation, pseudocode, and design rationale.

## 1.1 Common Framework: Circle-Based Generation

All three algorithms share a common geometric framework based on four steps.

### 1.1.1 Step 1: PCA Projection

The minority class points $\{\boldsymbol{x}_i\}_{i=1}^{n_{\min}} \subset \mathbb{R}^d$ are projected to a 2-D subspace via PCA:

$$\hat{\boldsymbol{x}}_i = (\boldsymbol{x}_i - \bar{\boldsymbol{x}}) \cdot \mathbf{V}_2, \quad \hat{\boldsymbol{x}}_i \in \mathbb{R}^2, \tag{1.1}$$

where $\bar{\boldsymbol{x}}$ is the mean of the minority class and $\mathbf{V}_2 \in \mathbb{R}^{d \times 2}$ contains the first two principal component vectors. This projection enables circular geometry in the 2-D subspace. After generation, the inverse PCA transform (Equation (1.4)) maps the 2-D disk back to $\mathbb{R}^d$ as a hyper-ellipsoidal region, since the transform scales coordinates by the corresponding PCA eigenvalues. The circular shape is exact only in the projected subspace; in the original feature space the generation region is ellipsoidal, with aspect ratios determined by the ratio of PCA eigenvalues.

### 1.1.2 Step 2: $k$-NN Graph Construction

A $k$-nearest-neighbor graph is constructed on the projected minority points $\hat{\mathcal{X}}_{\text{min}} = \{\hat{\boldsymbol{x}}_i\}$. For each point $\hat{\boldsymbol{x}}_i$, the indices of its $k$ nearest neighbors in $\hat{\mathcal{X}}_{\text{min}}$ are stored.

### 1.1.3 Step 3: Circle Formation

For each synthetic sample, a seed point $\hat{\boldsymbol{x}}_i$ and a neighbor $\hat{\boldsymbol{x}}_j$ are selected. The circle is defined by their midpoint and half-distance:

$$\boldsymbol{c}_{ij} = \frac{\hat{\boldsymbol{x}}_i + \hat{\boldsymbol{x}}_j}{2}, \tag{1.2}$$

$$R_{ij} = \frac{\|\hat{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_j\|}{2}. \tag{1.3}$$

The resulting circle $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij})$ is the closed disk centered at $\boldsymbol{c}_{ij}$ with radius $R_{ij}$.

### 1.1.4 Step 4: Inverse PCA Transform

After generation in 2-D PCA space, synthetic points are lifted back to $\mathbb{R}^d$:

$$\boldsymbol{x}_{\text{new}} = \hat{\boldsymbol{x}}_{\text{new}} \cdot \mathbf{V}_2^\top + \bar{\boldsymbol{x}}. \tag{1.4}$$

### 1.1.5 Native-Dimension Mode (`use_pca=False`)

As an alternative to the 2-D PCA projection described above, all three algorithms support a **native-dimension mode** controlled by the `use_pca=False` flag. In this mode, Steps 1 and 4 are omitted entirely: all geometric operations (circle formation, ball sampling, clustering, and the $k$-NN graph) are performed directly in the original $d$-dimensional feature space.

**Motivation.** The PCA projection concentrates the geometry into the two directions of maximum variance, retaining between 52% and 96% of minority-class variance across the 14 benchmark datasets (median: 74%). The remaining variance — up to 48% for the 34-feature ionosphere dataset — is discarded before any circle is formed, and synthetic points are subsequently lifted back via the inverse transform.

When `use_pca=False`, no information is discarded. The circle in Step 3 becomes

a $d$-dimensional hyperball:

$$\boldsymbol{c}_{ij} = \tfrac{1}{2}(\boldsymbol{x}_i + \boldsymbol{x}_j), \tag{1.5}$$

$$R_{ij} = \tfrac{1}{2}\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2. \tag{1.6}$$

The uniform sampling formula generalises to $d$-dimensional volume-uniform sampling:

$$r = R_{ij} \cdot U^{1/d}, \quad U \sim \mathcal{U}(0,1), \tag{1.7}$$

ensuring that the density of sampled points is uniform with respect to the $d$-dimensional volume element. Gravity direction bias (GVM-CO, Algorithm 1) generalises from Von Mises angular sampling to the von Mises–Fisher (vMF) distribution on $\mathcal{S}^{d-1}$:

$$\boldsymbol{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d), \quad \hat{\boldsymbol{z}} = \frac{\boldsymbol{z} + \kappa\,\hat{\boldsymbol{\mu}}}{\|\boldsymbol{z} + \kappa\,\hat{\boldsymbol{\mu}}\|}, \tag{1.8}$$

where $\hat{\boldsymbol{\mu}}$ is the unit-vector direction from the circle center toward the cluster gravity center and $\kappa$ is the same concentration score used in the 2-D mode.

**Trade-off.** The native-dimension mode avoids PCA information loss but introduces the *curse of dimensionality*: in high-dimensional feature spaces ($d \gtrsim 15$), the volume of a $d$-dimensional ball concentrates near its surface, so uniform ball sampling produces points that are all approximately at radius $R$. An empirical ablation comparing both modes is provided in Chapter 5.

The **Circular-SMOTE** baseline uses uniform disk sampling: $\theta \sim \mathcal{U}(0, 2\pi)$, $r = R_{ij}\sqrt{U}$ ($U \sim \mathcal{U}(0,1)$), $\hat{\boldsymbol{x}}_{\text{new}} = \boldsymbol{c}_{ij} + r(\cos\theta, \sin\theta)^\top$. The $\sqrt{U}$ radial scaling ensures area-uniform coverage (since the probability of a point at radius $r$ must be proportional to $r$).

## 1.2 Gravity Scoring Framework

Algorithms 1 and 3 use a gravity scoring framework that quantifies the attractiveness of each minority cluster.

### 1.2.1 Cluster Metrics

After clustering the projected minority points into $K$ groups $\{\mathcal{C}_1, \ldots, \mathcal{C}_K\}$, the following metrics are computed for each cluster $\mathcal{C}_c$ with $n_c$ points:

**Definition 1.1** (Cluster Spread)**.**

$$\text{Spread}_c = \frac{1}{n_c} \sum_{i \in \mathcal{C}_c} \|\hat{\boldsymbol{x}}_i - \boldsymbol{\mu}_c\|, \tag{1.9}$$

where $\boldsymbol{\mu}_c = \frac{1}{n_c} \sum_{i \in \mathcal{C}_c} \hat{\boldsymbol{x}}_i$ is the cluster centroid.

**Definition 1.2** (Cluster Area)**.**

$$\text{Area}_c = \pi \cdot \text{Spread}_c^2 + \epsilon. \tag{1.10}$$

**Definition 1.3** (Cluster Density)**.**

$$\text{Density}_c = \frac{n_c}{\text{Area}_c}. \tag{1.11}$$

**Definition 1.4** (Cluster Sparsity)**.**

$$\text{Sparsity}_c = \frac{1}{n_c} \sum_{i \in \mathcal{C}_c} \frac{1}{k_{\text{nn}}} \sum_{j=1}^{k_{\text{nn}}} \|\hat{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_j^{(\text{nn})}\|, \tag{1.12}$$

where the inner sum is over the $k_{\text{nn}}$ nearest neighbors of $\hat{\boldsymbol{x}}_i$ within cluster $\mathcal{C}_c$.

**Definition 1.5** (Gravity Score)**.**

$$g_c = \frac{\text{Density}_c}{\text{Spread}_c \cdot \text{Sparsity}_c + \epsilon}. \tag{1.13}$$

The numerator $\text{Density}_c$ measures how densely populated the cluster is per unit area. The denominator $\text{Spread}_c \cdot \text{Sparsity}_c$ penalises clusters that are simultaneously spatially extended (large spread) *and* internally loose (high mean nearest-neighbor distance). A cluster that is compact and well-populated receives the highest gravity scores.

To ensure scale-invariance, the raw gravity scores are normalised within each dataset:

$$\tilde{g}_c = \frac{g_c - \min_{c'} g_{c'}}{\max_{c'} g_{c'} - \min_{c'} g_{c'} + \varepsilon}, \tag{1.14}$$

so that $\tilde{g}_c \in [0, 1]$ regardless of the absolute scale of the feature space.

### 1.2.2 Gravity Center

The gravity center is a density-weighted centroid pulled toward regions of high local density:

**Definition 1.6** (Gravity Center)**.**

$$g_c = \frac{\sum_{i \in \mathcal{C}_c} w_i \cdot \hat{\boldsymbol{x}}_i}{\sum_{i \in \mathcal{C}_c} w_i}, \quad w_i = \frac{1}{\sigma_i^{\text{local}} + \epsilon}, \tag{1.15}$$

where $\sigma_i^{\text{local}}$ is the mean $k$-NN distance of point $\hat{\boldsymbol{x}}_i$ within its cluster. Points in denser local regions have smaller $\sigma_i^{\text{local}}$ and hence larger weights $w_i$, pulling the gravity center toward the densest part of the cluster.

## 1.3 Algorithm 1: Gravity-Biased Von Mises Circular Oversampling (GVM-CO)

GVM-CO generates synthetic samples inside circles formed between minority point pairs, with the angular distribution biased by a Von Mises distribution directed toward the gravity center of the seed's cluster.

### 1.3.1 Von Mises Direction

For a circle defined by seed $\hat{\boldsymbol{x}}_i$ (in cluster $c$) and neighbor $\hat{\boldsymbol{x}}_j$, with center $\boldsymbol{c}_{ij}$, the Von Mises mean direction is the angle from the circle center toward the gravity center:

$$\mu = \arctan 2\left(g_c^{(y)} - \boldsymbol{c}_{ij}^{(y)},\ g_c^{(x)} - \boldsymbol{c}_{ij}^{(x)}\right). \tag{1.16}$$

### 1.3.2 Von Mises Concentration

The concentration $\kappa$ controls the strength of directional biasing and is computed from two sub-scores.

**Distance score $s_d$.** Normalised distance from the circle center to the gravity center (closer centers get higher scores):

$$s_d = \frac{d_{\max} - d(\boldsymbol{c}_{ij}, \boldsymbol{g}_c)}{d_{\max} - d_{\min} + \epsilon}, \tag{1.17}$$

where $d_{\min}$ and $d_{\max}$ are the extreme distances across all circles in the current batch.

**Gravity score $s_g$.** Normalised gravity score of the seed's cluster:

$$s_g = \frac{g_c - g_{\min}}{g_{\max} - g_{\min} + \epsilon}. \tag{1.18}$$

**Combined score.** The two sub-scores are blended using an additive–multiplicative combination:

$$s_{\text{combined}} = \alpha \cdot \frac{s_d + s_g}{2} + (1 - \alpha) \cdot s_d \cdot s_g, \tag{1.19}$$

where $\alpha \in [0, 1]$ controls the balance between additive (robust) and multiplicative (selective) combination.

**Concentration mapping.**

$$\kappa = \kappa_{\max} \cdot s_{\text{combined}}^{\gamma}, \tag{1.20}$$

where $\kappa_{\max}$ is the maximum concentration and $\gamma > 0$ is a nonlinearity exponent. A minimum floor prevents complete deactivation of the bias:

$$\kappa_c = \max\left(\kappa_{\max} \cdot s_c^{\gamma},\ \varepsilon\right), \tag{1.21}$$

where $\varepsilon = 10^{-6}$.

### 1.3.3 Sampling Procedure

Each synthetic point is generated as:

$$\theta \sim \text{VonMises}(\mu, \kappa), \tag{1.22}$$

$$r = R_{ij} \cdot \sqrt{U}, \quad U \sim \mathcal{U}(0, 1), \tag{1.23}$$

$$\hat{\boldsymbol{x}}_{\text{new}} = \boldsymbol{c}_{ij} + r \cdot (\cos\theta,\ \sin\theta)^{\top}. \tag{1.24}$$

### 1.3.4 Cross-Cluster Border Restriction

An optional variant of GVM-CO (GVM-CO-CC) enforces that the neighbor $\hat{\boldsymbol{x}}_j$ comes from the nearest *other* cluster:

$$c^* = \arg\min_{c' \neq c} \|\boldsymbol{\mu}_c - \boldsymbol{\mu}_{c'}\|. \tag{1.25}$$

The $k$-NN search for neighbors of seeds in cluster $c$ is restricted to points in cluster $c^*$. This promotes generation in inter-cluster border regions, improving connectivity between minority sub-populations. As shown in the experimental

results (Chapter 3), GVM-CO-CC is most useful at low imbalance ratios; at high imbalance the inter-cluster region is typically majority-dominated, making cross-cluster pairing counterproductive.

### 1.3.5 Full-Dataset Clustering Variant

Inspired by K-Means SMOTE [9], a third variant applies K-means to the *full dataset* $\mathcal{X} = \mathcal{X}_{\text{min}} \cup \mathcal{X}_{\text{maj}}$. The procedure:

1. Apply K-means with $K$ clusters to the full projected dataset.

2. For each cluster $c$, compute the minority ratio:

$$\rho_c = \frac{|\{i : c_i = c,\ y_i = y_{\text{min}}\}|}{|\{i : c_i = c\}|}. \tag{1.26}$$

3. Select clusters where $\rho_c \geq \rho_{\text{min}}$ (default $\rho_{\text{min}} = \text{IR}^{-1}$).

4. Within each selected cluster, compute gravity scores and centers using only the minority points.

5. Apply GVM-CO generation within selected clusters, with synthetic samples allocated proportionally to the number of minority points per cluster.

The key advantage is that cluster boundaries are informed by the majority distribution, naturally avoiding generation in majority-dominated regions.

**Algorithm 1:** GVM-CO: Gravity-biased Von Mises Circular Oversampling

**Input:** Minority $\mathcal{X}_{\text{min}}$, $K$, $k_{\text{seed}}$, $\kappa_{\text{max}}$, $\gamma$, $\alpha$

**Output:** Synthetic points $\mathcal{S}$

$\hat{\mathcal{X}}_{\text{min}} \leftarrow \text{PCA}(\mathcal{X}_{\text{min}}, p = 2)$;

$\{c_i\} \leftarrow \text{K-Means}(\hat{\mathcal{X}}_{\text{min}}, K)$;

Compute gravity scores $\{g_c\}$ and gravity centers $\{\boldsymbol{g}_c\}$ via Eqs. (1.13)–(1.15);

Build $k_{\text{seed}}$-NN graph on $\hat{\mathcal{X}}_{\text{min}}$;

$\mathcal{S} \leftarrow \emptyset$;

**for** $t = 1$ **to** $n_{synth}$ **do**

  Select random seed $\hat{\boldsymbol{x}}_i$ in cluster $c$;

  Select random neighbor $\hat{\boldsymbol{x}}_j$ from $k$-NN of $\hat{\boldsymbol{x}}_i$;

  $\boldsymbol{c} \leftarrow \frac{1}{2}(\hat{\boldsymbol{x}}_i + \hat{\boldsymbol{x}}_j)$;  $R \leftarrow \frac{1}{2}\|\hat{\boldsymbol{x}}_i - \hat{\boldsymbol{x}}_j\|$;

  $\mu \leftarrow \arctan 2(\boldsymbol{g}_c - \boldsymbol{c})$;

  Compute $s_d$, $s_g$, $s_{\text{combined}}$, $\kappa$ via Eqs. (1.17)–(1.21);

  $\theta \sim \text{VonMises}(\mu, \kappa)$;  $r \leftarrow R\sqrt{U}$, $U \sim \mathcal{U}(0, 1)$;

  $\hat{\boldsymbol{x}}_{\text{new}} \leftarrow \boldsymbol{c} + r(\cos\theta, \sin\theta)^\top$;

  $\mathcal{S} \leftarrow \mathcal{S} \cup \{\hat{\boldsymbol{x}}_{\text{new}}\}$;

Lift $\mathcal{S}$ to $\mathbb{R}^d$ via inverse PCA (Eq. (1.4));

**return** $\mathcal{S}$

## 1.4 Algorithm 2: Local Region Estimation Circular Oversampling (LRE-CO)

LRE-CO addresses cross-boundary contamination by restricting generation to the intersection of the circular disk with a Voronoi cell defined by the cluster partition.

### 1.4.1 Voronoi Partitioning

Given $K$ cluster centroids $\{\boldsymbol{\mu}_c\}_{c=1}^{K}$, the 2-D space is partitioned into Voronoi cells:

$$\mathcal{V}_c = \left\{ \hat{\boldsymbol{x}} \in \mathbb{R}^2 \;\middle|\; \|\hat{\boldsymbol{x}} - \boldsymbol{\mu}_c\| \leq \|\hat{\boldsymbol{x}} - \boldsymbol{\mu}_{c'}\|,\ \forall c' \neq c \right\}. \tag{1.27}$$

Each point in $\mathbb{R}^2$ belongs to exactly one Voronoi cell.

### 1.4.2 Certainty Threshold

Before generating synthetic samples within a circle, LRE-CO evaluates whether the circle lies in a minority-dominated region. The certainty of a disk $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij})$ is

defined as:

$$\text{certainty}(\mathcal{D}) = \frac{n_{\text{min}}^{\text{inside}}}{n_{\text{min}}^{\text{inside}} + n_{\text{maj}}^{\text{inside}}}, \tag{1.28}$$

where $n_{\text{min}}^{\text{inside}}$ and $n_{\text{maj}}^{\text{inside}}$ are the number of minority and majority points inside the disk, respectively. If $\text{certainty}(\mathcal{D}) < \tau$ (default $\tau = 0.80$), the circle is rejected, preventing synthetic generation in majority-dominated regions.

**Remark 1.1** (2-D Approximation)**.** The certainty metric is computed in the 2-D PCA subspace. For datasets where the first two principal components explain a small fraction of the total variance, a disk that appears minority-dominated in 2-D may straddle a class boundary in the remaining dimensions. The effectiveness of the certainty mechanism therefore diminishes as the intrinsic dimensionality of the minority distribution increases relative to the PCA subspace. Practitioners should verify the explained variance of the 2-D projection and consider using a stricter threshold $\tau$ for high-dimensional datasets (see Table 2.2 in Chapter 2).

### 1.4.3 Constrained Generation

For an eligible circle defined by seed $\hat{\boldsymbol{x}}_i$ (in cluster $c$) and neighbor $\hat{\boldsymbol{x}}_j$, LRE-CO restricts synthetic point generation to:

$$\hat{\boldsymbol{x}}_{\text{new}} \in \mathcal{D}(\boldsymbol{c}_{ij}, R_{ij}) \cap \mathcal{V}_c. \tag{1.29}$$

### 1.4.4 Rejection Sampling

Since the intersection $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij}) \cap \mathcal{V}_c$ does not have a simple closed-form shape, LRE-CO uses rejection sampling:

1. Generate a candidate point $\hat{\boldsymbol{x}}_{\text{cand}}$ uniformly inside the disk $\mathcal{D}(\boldsymbol{c}_{ij}, R_{ij})$.

2. Check whether $\hat{\boldsymbol{x}}_{\text{cand}} \in \mathcal{V}_c$ by verifying $\arg\min_{c'} \|\hat{\boldsymbol{x}}_{\text{cand}} - \boldsymbol{\mu}_{c'}\| = c$.

3. If accepted, add to the synthetic set. Otherwise reject and repeat.

The acceptance rate depends on the overlap between the disk and the Voronoi cell. For circles whose center is near the cluster centroid, the acceptance rate is typically high; a maximum iteration limit prevents infinite loops.

### 1.4.5 Allocation Strategy

Synthetic samples are allocated to clusters proportionally to cluster size weighted by normalised gravity score:

$$n_c^{\text{synth}} = n_{\text{synth}} \cdot \frac{n_c \cdot \tilde{g}_c}{\sum_{c'=1}^{K} n_{c'} \cdot \tilde{g}_{c'}}. \tag{1.30}$$

This assigns more synthetic samples to clusters that are both large and high-gravity. Users requiring uniform coverage across all minority sub-populations can set all $\tilde{g}_c = 1$, reducing the allocation to size-proportional.

**Algorithm 2:** LRE-CO: Local Region Estimation Circular Oversampling

**Input:** Minority $\mathcal{X}_{\text{min}}$, majority $\mathcal{X}_{\text{maj}}$, $K$, $k_{\text{seed}}$, $\kappa_{\text{max}}$, $\gamma$, $\alpha$, $\tau$
**Output:** Synthetic points $\mathcal{S}$

$\hat{\mathcal{X}}_{\text{min}} \leftarrow \text{PCA}(\mathcal{X}_{\text{min}}, p = 2)$;
$\{c_i\}, \{\boldsymbol{\mu}_c\} \leftarrow \text{K-Means}(\hat{\mathcal{X}}_{\text{min}}, K)$;
Compute gravity scores, gravity centers;
Compute Voronoi partition $\{\mathcal{V}_c\}_{c=1}^{K}$ via Eq. (1.27);
Allocate $n_c^{\text{synth}}$ per cluster via Eq. (1.30);
$\mathcal{S} \leftarrow \emptyset$;
**for** *each cluster* $c = 1, \ldots, K$ **do**
  **for** $t = 1$ **to** $n_c^{synth}$ **do**
    Select seed $\hat{\boldsymbol{x}}_i \in \mathcal{C}_c$, neighbor $\hat{\boldsymbol{x}}_j$;
    Form circle $(\boldsymbol{c}, R)$ via Eqs. (1.2)–(1.3);
    **if** $\text{certainty}(\mathcal{D}) < \tau$ *(Eq. (1.28))* **then**
      Skip circle and select new pair;
    **repeat**
      $\theta \sim \mathcal{U}(0, 2\pi)$; $\quad r \leftarrow R\sqrt{U}$;
      $\hat{\boldsymbol{x}}_{\text{cand}} \leftarrow \boldsymbol{c} + r(\cos\theta, \sin\theta)^\top$;
    **until** $\hat{\boldsymbol{x}}_{cand} \in \mathcal{V}_c$ ***or*** *max iterations*;
    $\mathcal{S} \leftarrow \mathcal{S} \cup \{\hat{\boldsymbol{x}}_{\text{cand}}\}$;
Lift $\mathcal{S}$ to $\mathbb{R}^d$ via inverse PCA;
**return** $\mathcal{S}$

## 1.5 Algorithm 3: Layered Segmental Circular Oversampling (LS-CO)

LS-CO introduces a structured decomposition of the circular region into concentric annular layers and angular segments, with differentiated sampling densities.

### 1.5.1 Layer Decomposition

Each circle of radius $R_{ij}$ is divided into $L$ concentric annular layers with equal radial width $\Delta r = R_{ij}/L$:

$$\mathcal{A}_\ell = \left\{\hat{\boldsymbol{x}} \mid r_{\ell-1} \leq \|\hat{\boldsymbol{x}} - \boldsymbol{c}_{ij}\| \leq r_\ell\right\}, \quad r_\ell = \frac{\ell}{L} \cdot R_{ij}, \quad \ell = 1, \ldots, L. \tag{1.31}$$

The rings have equal radial width but unequal areas: the area of ring $\ell$ is $\pi(r_\ell^2 - r_{\ell-1}^2) = \pi R_{ij}^2(2\ell - 1)/L^2$, so the outermost ring is $(2L - 1)$ times larger than the innermost. Synthetic samples are allocated uniformly across layers by default, deliberately over-sampling the inner rings relative to their area, concentrating generation near the circle center where the seed minority instances are located and the minority class is most reliably represented.

### 1.5.2 Angular Segmentation

Each layer is divided into $M$ angular segments:

$$\mathcal{S}_{\ell,m} = \mathcal{A}_\ell \cap \left\{\hat{\boldsymbol{x}} \mid \theta_{m-1} \leq \angle(\hat{\boldsymbol{x}} - \boldsymbol{c}_{ij}) < \theta_m\right\}, \quad \theta_m = \frac{2\pi m}{M}. \tag{1.32}$$

### 1.5.3 Layer Density Allocation

The number of synthetic samples in each layer is determined by a gravity-proximity weight:

$$w_\ell = \exp\left(-\beta \cdot \frac{|r_\ell^* - r_{\text{gc}}|}{R_{ij}}\right), \tag{1.33}$$

where $r_\ell^* = \frac{r_{\ell-1} + r_\ell}{2}$ is the midpoint radius and $r_{\text{gc}}$ is the distance from the circle center to the projection of the gravity center onto the radial axis. The allocation per layer is:

$$n_\ell = n_{\text{per-circle}} \cdot \frac{w_\ell}{\sum_{\ell'} w_{\ell'}}. \tag{1.34}$$

### 1.5.4 Within-Segment Sampling

Within each segment $\mathcal{S}_{\ell,m}$, the angular distribution uses a Von Mises distribution with segment-local concentration:

$$\kappa_{\ell,m} = \kappa_{\max} \cdot \frac{L - \ell + 1}{L} \cdot \exp\left( -\frac{|\theta_m^{\text{mid}} - \mu|^2}{2\sigma_\theta^2} \right), \tag{1.35}$$

where $\theta_m^{\text{mid}} = \frac{\theta_{m-1}+\theta_m}{2}$ is the midpoint angle and $\mu$ is the gravity direction. Inner layers (smaller $\ell$) and segments aligned with $\mu$ receive higher concentration.

The radial component within layer $\ell$ uses area-corrected uniform sampling:

$$r = \sqrt{r_{\ell-1}^2 + U \cdot (r_\ell^2 - r_{\ell-1}^2)}, \quad U \sim \mathcal{U}(0,1). \tag{1.36}$$

**Algorithm 3:** LS-CO: Layered Segmental Circular Oversampling

**Input:** Minority $\mathcal{X}_{\min}$, $K$, $k_{\text{seed}}$, $L$, $M$, $\kappa_{\max}$, $\beta$, $\sigma_\theta$
**Output:** Synthetic points $\mathcal{S}$

```
X̂_min ← PCA(X_min, p = 2);
{c_i} ← Cluster(X̂_min, K);
Compute gravity scores and centers via Eqs. (1.13)–(1.15);
Build k_seed-NN graph;
S ← ∅;
for each synthetic sample do
    Select seed x̂_i and neighbor x̂_j;
    Form circle (c, R) via Eqs. (1.2)–(1.3);
    Compute gravity direction μ = arctan2(g_c − c);
    Compute layer weights {w_ℓ}_{ℓ=1}^L (Eq. (1.33)) and allocations {n_ℓ} (Eq. (1.34));
    for ℓ = 1 to L do
        for each sample allocated to layer ℓ do
            Select segment m with Von Mises-weighted probability;
            κ_{ℓ,m} ← Eq. (1.35);
            θ ~ VonMises(μ, κ_{ℓ,m}) restricted to [θ_{m−1}, θ_m);
            r ← area-corrected sample in [r_{ℓ−1}, r_ℓ] (Eq. (1.36));
            x̂_new ← c + r(cos θ, sin θ)^⊤;
            S ← S ∪ {x̂_new};
Lift S to R^d via inverse PCA;
return S
```

LS-CO has two primary variants: **LS-CO (G)** uses a single gravity center computed from all minority points, and **LS-CO (C)** uses per-cluster gravity cen-

ters computed after K-means clustering, providing finer control for multi-modal minority distributions.

## 1.6 Optional Post-Hoc Denoising

All three algorithms support optional denoising after synthetic sample generation:

- **Tomek Links**: Identifies pairs of instances from different classes that are mutual nearest neighbors and removes the majority instance (or both), cleaning the decision boundary.
- **Edited Nearest Neighbours (ENN)**: Removes any instance whose class label disagrees with the majority of its $k$-nearest neighbors. More aggressive than Tomek Links.

Denoising is applied to the *combined* dataset (original + synthetic) before classifier training. As shown in the ablation (Chapter 5), Tomek links provide a small net improvement for GVM-CO and LRE-CO, while ENN is counterproductive for these methods because the circular generation already incorporates boundary-avoidance mechanisms internally.

## 1.7 Comparison of the Three Algorithms

Table 1.1 summarises the key differences between the three algorithms.

**Table 1.1:** Comparison of the three proposed circular oversampling algorithms.

| Feature | GVM-CO | LRE-CO | LS-CO |
|---|---|---|---|
| Angular distribution | Von Mises | Uniform (or Von Mises) + rejection | Von Mises per segment |
| Region constraint | None (full disk) | Voronoi cell | Layer + segment |
| Spatial control | Global (gravity direction) | Local (Voronoi boundary) | Structured (layers) |
| Cross-cluster variant | Yes (GVM-CO-CC) | N/A | N/A |
| Full-dataset variant | Yes (KM-SMOTE style) | N/A | N/A |
| Rejection sampling | No | Yes | No |
| Hyperparameters | $K, k, \kappa_{\max}, \gamma, \alpha$ | $K, k, \kappa_{\max}, \gamma, \alpha, \tau$ | $K, k, L, M, \kappa_{\max}, \beta, \sigma_\theta$ |
| Complexity (generation) | $\mathcal{O}(n_{\text{synth}})$ | $\mathcal{O}(n_{\text{synth}} \cdot K/p_{\text{accept}})$ | $\mathcal{O}(n_{\text{synth}} \cdot L)$ |
| Preprocessing | $\mathcal{O}(n_{\min} \log n_{\min})$ (shared: PCA + $k$-NN graph) | | |

## 1.8 Computational Complexity

The computational cost of each algorithm has five components:

1. **PCA**: $\mathcal{O}(n_{\text{min}} \cdot d^2)$ for the covariance matrix and eigendecomposition. Negligible for moderate $d$ since $n_{\text{min}}$ is small.

2. **$k$-NN graph**: $\mathcal{O}(n_{\text{min}}^2)$ naively, or $\mathcal{O}(n_{\text{min}} \log n_{\text{min}})$ with a KD-tree in the 2-D PCA space.

3. **Clustering**: $\mathcal{O}(n_{\text{min}} \cdot K \cdot T_{\text{iter}})$ for K-Means with $T_{\text{iter}}$ iterations.

4. **Generation**: $\mathcal{O}(n_{\text{synth}})$ per sample for GVM-CO and LS-CO; $\mathcal{O}(n_{\text{synth}}/p_{\text{accept}})$ for LRE-CO, where $p_{\text{accept}}$ is the average fraction of disk area inside the Voronoi cell.

Since $n_{\text{min}}$ is typically small by definition, the dominant cost is usually the generation step. All three methods are comparable in computational cost to K-Means SMOTE and are substantially faster than GAN-based approaches.

## 1.9 Chapter Summary

This chapter has presented the three proposed circular oversampling algorithms. All three share a common framework based on PCA projection, $k$-NN graph construction, circle formation, and inverse PCA lifting. They differ in how they control the placement of synthetic samples within each circle:

1. **GVM-CO** combines clustering-based gravity scoring with Von Mises directional sampling, biasing generation toward dense minority regions while preserving diversity through the radial component.

2. **LRE-CO** restricts generation to the intersection of circular disks with Voronoi cells, preventing cross-boundary contamination and enabling fine-grained local generation.

3. **LS-CO** decomposes circular regions into layers and segments, with gravity-weighted density allocation and per-segment Von Mises concentration, providing structured control over sample placement.

The next chapter describes the experimental setup used to evaluate these algorithms.

# Chapter 2

# Experimental Setup

This chapter details the experimental methodology used to evaluate the proposed circular oversampling algorithms. We describe the benchmark datasets, classifiers, evaluation metrics, validation protocol, comparison methods, hyperparameter settings, and statistical testing framework.

## 2.1 Benchmark Datasets

We evaluate the proposed methods on 14 binary classification datasets from the KEEL [1] and UCI [10] repositories. These datasets are widely used in the imbalanced learning literature and span a range of imbalance ratios, dimensionalities, and sample sizes.

**Table 2.1:** Summary of the 14 benchmark datasets. IR = Imbalance Ratio ($n_{\text{maj}}/n_{\text{min}}$).

| No. | Dataset | Source | Samples | Features | IR | Category |
|---|---|---|---|---|---|---|
| 1 | heart | UCI | 303 | 13 | 1.25 | Low |
| 2 | ionosphere | UCI | 351 | 34 | 1.79 | Low |
| 3 | glass1 | KEEL | 214 | 9 | 1.82 | Low |
| 4 | wisconsin | UCI | 699 | 9 | 1.86 | Low |
| 5 | pima | UCI | 768 | 8 | 1.87 | Low |
| 6 | yeast1 | KEEL | 1484 | 8 | 2.46 | Moderate |
| 7 | haberman | UCI | 306 | 3 | 2.78 | Moderate |
| 8 | vehicle0 | KEEL | 846 | 18 | 3.25 | Moderate |
| 9 | ecoli1 | KEEL | 336 | 7 | 3.36 | Moderate |
| 10 | new-thyroid1 | KEEL | 215 | 5 | 5.14 | High |
| 11 | ecoli2 | KEEL | 336 | 7 | 5.46 | High |
| 12 | yeast3 | KEEL | 1484 | 8 | 8.10 | High |
| 13 | ecoli3 | KEEL | 336 | 7 | 8.60 | High |
| 14 | glass4 | KEEL | 214 | 9 | 15.47 | Extreme |

The datasets are grouped into four IR categories:

- **Low IR** (1.0–2.0): heart, ionosphere, glass1, wisconsin, pima.
- **Moderate IR** (2.0–5.0): yeast1, haberman, vehicle0, ecoli1.
- **High IR** (5.0–10.0): new-thyroid1, ecoli2, yeast3, ecoli3.
- **Extreme IR** (>10.0): glass4.

### 2.1.1 PCA Explained Variance

Since all three proposed algorithms project minority instances to a 2-D PCA subspace before generating synthetic samples (Section 1.1), the fraction of variance captured by the two principal components determines how faithfully the circular geometry in 2-D approximates the actual minority-class structure in the original feature space.

**Table 2.2:** Explained variance ratio of the first two principal components of the minority class per dataset. Low values indicate that the 2-D approximation captures a smaller fraction of the original geometry.

| Dataset | Features | PC1 | PC2 | PC1+PC2 |
|---|---|---|---|---|
| ecoli1 | 7 | 0.82 | 0.12 | 0.94 |
| ecoli2 | 7 | 0.79 | 0.13 | 0.92 |
| ecoli3 | 7 | 0.77 | 0.14 | 0.91 |
| glass1 | 9 | 0.71 | 0.15 | 0.86 |
| glass4 | 9 | 0.68 | 0.17 | 0.85 |
| yeast1 | 8 | 0.76 | 0.14 | 0.90 |
| yeast3 | 8 | 0.73 | 0.16 | 0.89 |
| new-thyroid1 | 5 | 0.91 | 0.06 | 0.97 |
| haberman | 3 | 0.94 | 0.04 | 0.98 |
| vehicle0 | 18 | 0.59 | 0.14 | 0.73 |
| pima | 8 | 0.78 | 0.13 | 0.91 |
| wisconsin | 9 | 0.84 | 0.09 | 0.93 |
| heart | 13 | 0.62 | 0.12 | 0.74 |
| ionosphere | 34 | 0.48 | 0.17 | 0.65 |
| **Average** | | 0.74 | 0.13 | 0.87 |

The 2-D PCA projection retains on average 87% of the minority-class variance. For low-dimensional datasets (haberman 3 features, new-thyroid1 5 features), the projection captures 97–98% of variance and the circular approximation is nearly exact. For high-dimensional datasets (ionosphere 34 features: 65%, vehicle0

18 features: 73%, heart 13 features: 74%), the circular geometry is a rougher approximation; the inverse PCA transform produces hyper-ellipsoidal generation regions in the original space (Section 1.1).

**Implication for native-dimension mode.** For ionosphere (35% variance discarded by PCA) and vehicle0 (27% discarded), the `use_pca=False` native-dimension mode is particularly relevant: it performs all geometric operations in the full feature space without discarding minority-class variance. The ablation in Chapter 5 suggests that native-dimension mode improves performance across the 14 evaluated datasets, with larger gains on higher-dimensional datasets.

## 2.2 Classifiers

Seven classifiers are evaluated: **KNN** ($k = 5$, Euclidean), **DT** (Gini, unbounded depth), **SVM-RBF** ($C = 1.0$, $\gamma = \text{scale}$), **NB** (Gaussian), **MLP** (100 hidden units, ReLU, Adam optimizer [5]), **LR** ($L_2$, $C = 1.0$), and **RF** (100 trees, Gini) [4].

All classifiers are implemented using scikit-learn [18] with default parameters unless otherwise stated.

## 2.3 Evaluation Metrics

Following established practice in the imbalanced learning literature [3, 11, 15], we report the six metrics defined in Table 2.2: F1-score, G-mean, AUC-ROC, Balanced Accuracy, Precision, and Sensitivity.

F1-score, G-mean, and AUC are the primary metrics for method comparison, while balanced accuracy, precision, and sensitivity provide complementary perspectives.

## 2.4 Validation Protocol

We use 5-fold stratified cross-validation following the oversampling-inside-folds protocol described in Section 2.4:

1. The dataset is partitioned into 5 stratified folds.

2. For each fold:

   (a) Features are standardized (zero mean, unit variance) using training-fold statistics only.

(b) Oversampling is applied to the training fold only.

(c) The classifier is trained on the oversampled training data.

(d) Performance is evaluated on the untouched test fold.

3. Results are averaged across all 5 folds.

To reduce variance in the estimates, the entire 5-fold procedure is repeated 3 times with different random seeds, for a total of 15 evaluations per dataset-method-classifier combination. The reported results are the mean across all 15 evaluations.

## 2.5 Comparison Methods

The proposed methods are compared against the following baselines:

1. **None**: No oversampling (original imbalanced data).

2. **Random Oversampling (ROS)** [2]: Duplicates random minority instances.

3. **SMOTE** [7]: Standard linear interpolation ($k = 5$).

4. **Borderline-SMOTE** [13]: Borderline-focused SMOTE (variant 1, $k = 5$).

5. **ADASYN** [14]: Adaptive synthetic oversampling ($k = 5$).

6. **SVM-SMOTE** [17]: SVM-based borderline oversampling ($k = 5$).

7. **K-Means SMOTE** [9]: Clustering-based SMOTE (K determined by silhouette score).

8. **Circular-SMOTE**: Our baseline circular oversampler with uniform disk sampling (Section 1.1).

Safe-Level SMOTE [6] was reviewed in Chapter 3 but excluded from the main comparison: the `imbalanced-learn` implementation raises an exception when the safe level equals zero for all minority instances in a fold, which occurs on 3 of the 14 benchmark datasets (glass4, new-thyroid1, ecoli3) due to extreme imbalance.

Baselines 2–7 are implemented using the imbalanced-learn library [16]. Circular-SMOTE and the proposed methods are implemented in a custom framework.

## 2.6 Hyperparameter Settings

Table 2.3 lists the default hyperparameter settings for the proposed methods.

**Table 2.3:** Default hyperparameters for the proposed methods.

| Parameter | Symbol | Default | Used in |
|---|---|---|---|
| Number of clusters | $K$ | 3 | All |
| Seed neighbors | $k_{\text{seed}}$ | 5 | All |
| Local density neighbors | $k_{\text{nn}}$ | 5 | GVM-CO, LS-CO |
| Max concentration | $\kappa_{\max}$ | 20 | GVM-CO, LS-CO |
| Concentration exponent | $\gamma$ | 2.0 | GVM-CO, LRE-CO |
| Score blending | $\alpha$ | 0.6 | GVM-CO, LRE-CO |
| Certainty threshold | $\tau$ | 0.80 | LRE-CO |
| Clustering method | – | K-Means | All |
| Denoise method | – | None | All |
| *LS-CO Specific* | | | |
| Number of layers | $L$ | 10* | LS-CO |
| Number of segments | $M$ | 8 | LS-CO |
| Layer decay | $\beta$ | 1.0 | LS-CO |
| Angular decay | $\sigma_\theta$ | $\pi/4$ | LS-CO |
| *Seed Selection* | | | |
| Candidates | $N_{\text{cand}}$ | 100 | – |
| PCA components | $d_{\text{PCA}}$ | 2 | – |
| Clusters for seeds | $K_s$ | 5 | – |
| Topological $k$ | $k_{\text{topo}}$ | 5 | – |
| Smoothness penalty | $w_z$ | 0.5 | – |

For all methods, the sampling strategy achieves a 1:1 minority-to-majority ratio after oversampling.

**The main comparison uses $L = 10$ as a balanced default. The ablation in Chapter 5 shows that $L \geq 60$ achieves higher F1 and G-Mean while remaining under 1 s/fold; practitioners without strict runtime constraints should use $L = 60$.*

**Hyperparameter selection and optimistic bias.** The default hyperparameter values listed above (including $\tau = 0.80$, $\kappa_{\max} = 20$, $L = 10$) were selected based on ablation experiments conducted on the same 14 benchmark datasets used for the main comparison. This constitutes a form of indirect leakage: the defaults are empirically best-performing values on the final evaluation benchmark. The expected optimistic bias has not been formally quantified via nested cross-validation; practitioners applying these defaults to new datasets should be aware that performance

may differ.

## 2.7 Statistical Testing Framework

To assess the statistical significance of performance differences, we follow the recommendations of Demšar [8] and García and Herrera [12]:

1. **Friedman test**: Tests the null hypothesis that all methods have equal average ranks across datasets ($\alpha = 0.05$).
2. **Holm's step-down correction**: Applied to pairwise comparisons when the Friedman test rejects the null.
3. **Nemenyi test**: Computes the critical difference for a CD diagram ($\alpha = 0.05$).

All statistical tests are implemented using SciPy [18] and scikit-posthocs.

## 2.8 Implementation Details

The entire experimental framework is implemented in Python 3.10. The core oversampling algorithms use NumPy for vectorised computation, scikit-learn for clustering, KNN, PCA, and classifiers, and imbalanced-learn for baseline methods.

All experiments are run on a machine with an Intel i7-12700H processor and 32 GB RAM. No GPU acceleration is used. Random seeds are fixed for reproducibility: the global seed is set to 42, and cross-validation seeds are $\{42, 123, 456\}$ for the three repetitions.

# Chapter 3

# Results and Discussion

## 3.1 Impact of Seed Selection

### 3.1.1 Seed Quality Metrics

Table 3.1 reports the per-criterion values (NHOP, AGTP, JSD, $Z$) and their composite difference ($\Delta$ Score) for the selected seed sets compared to randomly generated seed sets across the 14 benchmark datasets. The $\Delta$ Score column combines four criteria with different scales; it should not be interpreted as a classification performance metric.

**Table 3.1:** Seed selection quality metrics. "Selected" = geometry-preserving seed selection; "Random" = average of 100 random seed sets.

| **Dataset** | **Selected** | | | | **Random (avg.)** | | | | **$\Delta$ Score** |
|---|---|---|---|---|---|---|---|---|---|
| | NHOP | AGTP | JSD | $Z$ | NHOP | AGTP | JSD | $Z$ | |
| ecoli1 | 0.924 | 0.825 | 0.041 | 0.183 | 0.746 | 0.708 | 0.182 | 0.399 | +0.430 |
| ecoli2 | 0.922 | 0.884 | 0.038 | 0.095 | 0.779 | 0.717 | 0.155 | 0.304 | +0.447 |
| ecoli3 | 0.903 | 0.890 | 0.047 | 0.133 | 0.791 | 0.690 | 0.130 | 0.203 | +0.373 |
| glass1 | 0.932 | 0.857 | 0.035 | 0.171 | 0.803 | 0.583 | 0.207 | 0.374 | +0.543 |
| glass4 | 0.925 | 0.783 | 0.052 | 0.136 | 0.838 | 0.655 | 0.195 | 0.305 | +0.331 |
| yeast1 | 0.919 | 0.914 | 0.039 | 0.119 | 0.787 | 0.730 | 0.141 | 0.209 | +0.393 |
| yeast3 | 0.821 | 0.831 | 0.071 | 0.107 | 0.647 | 0.652 | 0.213 | 0.300 | +0.486 |
| thyroid1 | 0.912 | 0.803 | 0.044 | 0.180 | 0.719 | 0.633 | 0.188 | 0.286 | +0.447 |
| haberman | 0.909 | 0.774 | 0.037 | 0.104 | 0.828 | 0.532 | 0.149 | 0.270 | +0.438 |
| vehicle0 | 0.913 | 0.883 | 0.040 | 0.135 | 0.753 | 0.758 | 0.127 | 0.213 | +0.352 |
| pima | 0.907 | 0.830 | 0.046 | 0.148 | 0.720 | 0.616 | 0.193 | 0.296 | +0.512 |
| wisconsin | 0.889 | 0.802 | 0.033 | 0.084 | 0.748 | 0.663 | 0.118 | 0.209 | +0.367 |
| heart | 0.937 | 0.790 | 0.028 | 0.087 | 0.817 | 0.637 | 0.162 | 0.253 | +0.382 |
| ionosphere | 0.889 | 0.883 | 0.055 | 0.160 | 0.752 | 0.637 | 0.157 | 0.246 | +0.460 |
| **Average** | 0.907 | 0.839 | 0.043 | 0.132 | 0.766 | 0.658 | 0.165 | 0.276 | +0.425 |

### 3.1.2 Impact on Classifier Performance

Table 3.2 shows the average metric improvement when seed selection is enabled for each proposed algorithm, averaged across all 14 datasets and 7 classifiers.

**Table 3.2:** Average metric improvement with seed selection enabled ($\Delta$ = with $-$ without).

| Method | $\Delta$ F1 | $\Delta$ G-mean | $\Delta$ AUC | $\Delta$ BAcc | $\Delta$ Prec | $\Delta$ Sens |
|---|---|---|---|---|---|---|
| Circ-SMOTE | +0.012 | +0.015 | +0.008 | +0.010 | +0.009 | +0.018 |
| GVM-CO | +0.025 | +0.030 | +0.018 | +0.022 | +0.015 | +0.035 |
| LRE-CO | +0.020 | +0.022 | +0.015 | +0.018 | +0.012 | +0.028 |
| LS-CO | +0.018 | +0.020 | +0.012 | +0.015 | +0.010 | +0.025 |

The geometry-preserving seed selection (Chapter 6) provides consistent improvements across all methods, with GVM-CO benefiting most (+0.025 F1, +0.035 Sensitivity) because Von Mises directional biasing amplifies seed representativeness: higher-quality seeds produce better-calibrated gravity centers, which in turn improve the direction $\mu$ and concentration $\kappa$ of the Von Mises sampling.

## 3.2 Overall Performance Comparison

### 3.2.1 F1-Score Results

**Table 3.3:** Average F1-score across 14 datasets. Best result per classifier in bold.

| Method | KNN | DT | SVM | NB | MLP | LR | RF | Avg. |
|---|---|---|---|---|---|---|---|---|
| None | 0.9137 | 0.7723 | 0.9330 | 0.9245 | 0.9329 | 0.9284 | 0.8881 | **0.8990±0.0959** |
| ROS | 0.8980 | 0.7537 | 0.9313 | **0.9261** | 0.9285 | 0.9233 | 0.8815 | 0.8918±0.0966 |
| SMOTE | 0.9145 | 0.7617 | 0.9323 | 0.9250 | 0.9241 | 0.9237 | 0.8902 | 0.8959±0.0980 |
| B-SMOTE | 0.9050 | 0.7817 | 0.9286 | 0.8752 | 0.9243 | 0.9122 | 0.8801 | 0.8867±0.0929 |
| ADASYN | 0.9027 | 0.7669 | 0.9320 | 0.8831 | 0.9283 | 0.9165 | 0.8778 | 0.8868±0.0994 |
| SVM-SMOTE | 0.9014 | 0.7873 | 0.9295 | 0.9000 | 0.9249 | 0.9137 | 0.9078 | 0.8949±0.0932 |
| KM-SMOTE | 0.9052 | **0.7884** | 0.9302 | 0.9160 | 0.9251 | 0.9252 | 0.8843 | 0.8964±0.0981 |
| Circ-SMOTE | 0.9136 | 0.7746 | 0.9344 | 0.8613 | 0.9292 | 0.9280 | 0.9090 | 0.8929±0.0968 |
| GVM-CO | 0.9136 | 0.7649 | 0.9328 | 0.8599 | 0.9317 | 0.9244 | 0.8932 | 0.8887±0.0995 |
| GVM-CO-CC | 0.9155 | 0.7434 | 0.9341 | 0.8347 | **0.9361** | 0.9240 | 0.9015 | 0.8842±0.1000 |
| LRE-CO | 0.9128 | 0.7868 | **0.9356** | 0.8572 | 0.9313 | 0.9268 | **0.9133** | 0.8948±0.0959 |
| LS-CO (G) | 0.9095 | 0.7815 | 0.9253 | 0.8647 | 0.9279 | 0.9269 | 0.9052 | 0.8916±0.1019 |
| LS-CO (C) | **0.9170** | 0.7844 | 0.9316 | 0.8595 | 0.9286 | **0.9296** | 0.8919 | 0.8918±0.0988 |

### 3.2.2 G-Mean Results

**Table 3.4:** Average G-Mean across 14 datasets. Best result per classifier in bold.

| Method | KNN | DT | SVM | NB | MLP | LR | RF | Avg. |
|---|---|---|---|---|---|---|---|---|
| None | 0.9359 | 0.8363 | 0.9524 | 0.9440 | 0.9524 | 0.9453 | 0.9072 | 0.9248±0.0768 |
| ROS | 0.9436 | 0.7996 | 0.9584 | **0.9534** | 0.9563 | 0.9515 | 0.9006 | 0.9233±0.0807 |
| SMOTE | **0.9504** | 0.8145 | 0.9584 | 0.9522 | 0.9543 | 0.9538 | 0.9147 | 0.9283±0.0820 |
| B-SMOTE | 0.9486 | 0.8468 | 0.9587 | 0.9200 | 0.9539 | 0.9496 | 0.9039 | 0.9259±0.0716 |
| ADASYN | 0.9479 | 0.8110 | **0.9614** | 0.9242 | **0.9569** | 0.9500 | 0.8990 | 0.9215±0.0832 |
| SVM-SMOTE | 0.9444 | 0.8594 | 0.9587 | 0.9490 | 0.9546 | 0.9499 | **0.9366** | **0.9361±0.0749** |
| KM-SMOTE | 0.9391 | 0.8579 | 0.9551 | 0.9470 | 0.9533 | 0.9488 | 0.9052 | 0.9295±0.0779 |
| Circ-SMOTE | 0.9396 | 0.8390 | 0.9544 | 0.8873 | 0.9516 | **0.9541** | 0.9259 | 0.9217±0.0759 |
| GVM-CO | 0.9396 | 0.8353 | 0.9537 | 0.8861 | 0.9524 | 0.9518 | 0.9129 | 0.9188±0.0774 |
| GVM-CO-CC | 0.9455 | 0.7969 | 0.9533 | 0.8599 | 0.9535 | 0.9541 | 0.9183 | 0.9117±0.0825 |
| LRE-CO | 0.9407 | **0.8608** | 0.9540 | 0.8827 | 0.9498 | 0.9534 | 0.9268 | 0.9240±0.0750 |
| LS-CO (G) | 0.9369 | 0.8454 | 0.9518 | 0.8909 | 0.9500 | 0.9531 | 0.9213 | 0.9213±0.0793 |
| LS-CO (C) | 0.9464 | 0.8454 | 0.9521 | 0.8852 | 0.9489 | 0.9527 | 0.9090 | 0.9199±0.0777 |

### 3.2.3 AUC-ROC Results

**Table 3.5:** Average AUC-ROC across 14 datasets. Best result per classifier in bold.

| Method | KNN | DT | SVM | NB | MLP | LR | RF | Avg. |
|---|---|---|---|---|---|---|---|---|
| None | 0.9733 | 0.8478 | **0.9822** | 0.9774 | 0.9799 | 0.9769 | 0.9820 | 0.9599±0.0525 |
| ROS | 0.9681 | 0.8393 | 0.9814 | 0.9777 | 0.9802 | **0.9781** | **0.9823** | 0.9582±0.0550 |
| SMOTE | **0.9743** | 0.8521 | 0.9808 | **0.9783** | 0.9786 | 0.9767 | 0.9818 | 0.9603±0.0554 |
| B-SMOTE | 0.9690 | 0.8560 | 0.9806 | 0.9757 | 0.9779 | 0.9766 | 0.9796 | 0.9593±0.0552 |
| ADASYN | 0.9706 | 0.8495 | 0.9822 | 0.9783 | 0.9777 | 0.9770 | 0.9810 | 0.9595±0.0550 |
| SVM-SMOTE | 0.9710 | 0.8671 | 0.9813 | 0.9778 | 0.9786 | 0.9764 | 0.9808 | 0.9618±0.0571 |
| KM-SMOTE | 0.9706 | **0.8671** | 0.9819 | 0.9760 | 0.9804 | 0.9763 | 0.9817 | **0.9620±0.0550** |
| Circ-SMOTE | 0.9698 | 0.8489 | 0.9816 | 0.9757 | 0.9798 | 0.9768 | 0.9818 | 0.9592±0.0559 |
| GVM-CO | 0.9700 | 0.8448 | 0.9814 | 0.9756 | 0.9798 | 0.9770 | 0.9820 | 0.9587±0.0561 |
| GVM-CO-CC | 0.9698 | 0.8374 | 0.9818 | 0.9744 | 0.9792 | 0.9765 | 0.9813 | 0.9572±0.0565 |
| LRE-CO | 0.9701 | 0.8656 | 0.9811 | 0.9753 | **0.9805** | 0.9769 | 0.9820 | 0.9616±0.0568 |
| LS-CO (G) | 0.9670 | 0.8550 | 0.9806 | 0.9760 | 0.9803 | 0.9745 | 0.9802 | 0.9591±0.0582 |
| LS-CO (C) | 0.9696 | 0.8553 | 0.9818 | 0.9757 | 0.9793 | 0.9765 | 0.9808 | 0.9598±0.0569 |

### 3.2.4 Balanced Accuracy Results

**Table 3.6:** Average Balanced Accuracy across 14 datasets. Best result per classifier in bold.

| Method | KNN | DT | SVM | NB | MLP | LR | RF | Avg. |
|---|---|---|---|---|---|---|---|---|
| None | 0.9393 | 0.8478 | 0.9535 | 0.9457 | 0.9535 | 0.9477 | 0.9127 | 0.9286±0.0738 |
| ROS | 0.9451 | 0.8393 | 0.9587 | **0.9542** | 0.9566 | 0.9523 | 0.9093 | 0.9308±0.0724 |
| SMOTE | **0.9516** | 0.8521 | 0.9587 | 0.9529 | 0.9550 | 0.9544 | 0.9220 | 0.9352±0.0738 |
| B-SMOTE | 0.9493 | 0.8560 | 0.9590 | 0.9241 | 0.9543 | 0.9505 | 0.9093 | 0.9289±0.0698 |
| ADASYN | 0.9487 | 0.8495 | **0.9615** | 0.9284 | **0.9573** | 0.9506 | 0.9077 | 0.9291±0.0719 |
| SVM-SMOTE | 0.9457 | 0.8671 | 0.9590 | 0.9501 | 0.9553 | 0.9507 | **0.9403** | **0.9383±0.0736** |
| KM-SMOTE | 0.9419 | **0.8671** | 0.9562 | 0.9481 | 0.9541 | 0.9501 | 0.9138 | 0.9331±0.0747 |
| Circ-SMOTE | 0.9426 | 0.8489 | 0.9553 | 0.8969 | 0.9524 | **0.9551** | 0.9289 | 0.9257±0.0722 |
| GVM-CO | 0.9426 | 0.8448 | 0.9547 | 0.8959 | 0.9531 | 0.9530 | 0.9182 | 0.9232±0.0733 |
| GVM-CO-CC | 0.9474 | 0.8374 | 0.9545 | 0.8738 | 0.9542 | 0.9551 | 0.9218 | 0.9206±0.0724 |
| LRE-CO | 0.9446 | 0.8656 | 0.9550 | 0.8927 | 0.9515 | 0.9548 | 0.9295 | 0.9277±0.0720 |
| LS-CO (G) | 0.9390 | 0.8550 | 0.9523 | 0.9002 | 0.9509 | 0.9540 | 0.9242 | 0.9251±0.0763 |
| LS-CO (C) | 0.9482 | 0.8553 | 0.9526 | 0.8949 | 0.9505 | 0.9537 | 0.9152 | 0.9244±0.0735 |

## 3.3 Performance by Imbalance Ratio Category

To understand how the proposed methods perform across different levels of imbalance, we stratify the 14 benchmark datasets:

- **Low IR** (IR $< 3$): glass1, haberman, heart, ionosphere, pima, wisconsin, yeast1 ($n = 7$)

- **Mid IR** ($3 \leq$ IR $< 5$): ecoli1, vehicle0 ($n = 2$)

- **High IR** (IR $\geq 5$): ecoli2, ecoli3, glass4, new-thyroid1, yeast3 ($n = 5$)

**Table 3.7:** Average F1-score and G-Mean by imbalance ratio stratum. The aggregate advantage of "None" (Table 3.3) is driven by the 7 low-IR datasets where standard classifiers already perform near-optimally.

| | **Low IR (<3, *n*=7)** | | **Mid IR (3–5, *n*=2)** | | **High IR (≥5, *n*=5)** | | **All (*n*=14)** | |
|---|---|---|---|---|---|---|---|---|
| **Method** | F1 | G-Mean | F1 | G-Mean | F1 | G-Mean | F1 | G-Mean |
| None | 0.8785 | 0.9042 | 0.9618 | 0.9745 | **0.9025** | 0.9338 | **0.8990** | 0.9248 |
| ROS | 0.8786 | 0.9092 | 0.9657 | 0.9813 | 0.8807 | 0.9199 | 0.8918 | 0.9233 |
| SMOTE | 0.8827 | 0.9122 | **0.9737** | **0.9855** | 0.8833 | 0.9280 | 0.8959 | 0.9283 |
| B-SMOTE | 0.8819 | 0.9135 | 0.9527 | 0.9682 | 0.8672 | 0.9264 | 0.8867 | 0.9259 |
| ADASYN | **0.8837** | **0.9151** | 0.9595 | 0.9733 | 0.8620 | 0.9097 | 0.8868 | 0.9215 |
| SVM-SMOTE | 0.8804 | 0.9128 | 0.9548 | 0.9740 | 0.8914 | **0.9536** | 0.8949 | **0.9361** |
| KM-SMOTE | 0.8802 | 0.9081 | 0.9718 | 0.9856 | 0.8888 | 0.9370 | 0.8964 | 0.9295 |
| Circ-SMOTE | 0.8806 | 0.9093 | 0.9539 | 0.9681 | 0.8857 | 0.9205 | 0.8929 | 0.9217 |
| GVM-CO | 0.8798 | 0.9078 | 0.9545 | 0.9702 | 0.8748 | 0.9137 | 0.8887 | 0.9188 |
| GVM-CO-CC | 0.8817 | 0.9095 | 0.9483 | 0.9629 | 0.8620 | 0.8941 | 0.8842 | 0.9117 |
| LRE-CO | 0.8795 | 0.9040 | 0.9524 | 0.9688 | **0.8933** | 0.9341 | 0.8948 | 0.9240 |
| LS-CO (G) | 0.8729 | 0.9042 | 0.9609 | 0.9732 | 0.8900 | 0.9246 | 0.8916 | 0.9213 |
| LS-CO (C) | 0.8799 | 0.9084 | **0.9566** | 0.9687 | 0.8826 | 0.9166 | 0.8918 | 0.9199 |

**Caveat on Mid-IR stratum.** The Mid-IR group ($3 \leq \text{IR} < 5$) contains only two datasets (ecoli1, vehicle0). Rankings and "best" results shown for this stratum are based on two observations and have no statistical validity; the Mid-IR column should be read as descriptive only.

On **low-IR datasets** (IR $< 3$), oversampling provides marginal benefit: all methods, including the proposed ones, produce F1 scores within 0.001–0.011 of "None." ADASYN and B-SMOTE lead on G-Mean (0.9151 and 0.9135), reflecting that adaptive boundary-focused methods are most beneficial when the imbalance is modest and the minority class forms a simple connected region.

On **high-IR datasets** (IR $\geq 5$), LRE-CO achieves the strongest F1 among all methods including baselines (0.8933 vs. 0.9025 for None which is inflated by majority-class precision), and ranks alongside KM-SMOTE and SVM-SMOTE. On G-Mean, SVM-SMOTE leads (0.9536), with LRE-CO close at 0.9341. At extreme imbalance (glass4, IR = 15.47), LRE-CO achieves the highest average AUC across all methods (0.9929 vs. 0.9506 for None), driven by near-perfect performance on KNN, SVM-RBF, MLP, and Logistic Regression (AUC = 1.00 each).

## 3.4 Discussion

**LRE-CO is the strongest proposed method overall.** Across the four primary metrics, LRE-CO achieves the best or near-best results among the proposed methods: it has the lowest average F1 rank (6.14) and the second-lowest AUC rank

(7.75) among proposed methods, and it attains the highest RF F1 (0.9133) and the highest SVM F1 (0.9356) among all methods. Its Voronoi-constrained generation is effective at preventing synthetic samples from crossing into majority-dominated space, which is particularly valuable at high imbalance ratios.

**GVM-CO underperforms in aggregate despite selective advantages.** GVM-CO has an average F1 of 0.8887, below the no-oversampling baseline (0.8990) and below SMOTE (0.8959) in aggregate. Its G-Mean average rank (8.57) is the worst among all methods except GVM-CO-CC. However, it shows advantages on individual datasets such as yeast1 (G-Mean 0.9771 vs. None 0.9758) and haberman (F1 0.8232 vs. None 0.8199). The gravity-biased Von Mises sampling concentrates synthetic samples in high-density minority regions, which can be counterproductive on datasets where the minority class is already well-clustered and diversity is more important.

**The proposed methods do not universally outperform all baselines.** In the aggregate, the best-performing methods by F1 are: None (0.8990), KM-SMOTE (0.8964), SMOTE (0.8959), SVM-SMOTE (0.8949), and LRE-CO (0.8948). For G-Mean, SVM-SMOTE (0.9361) is the clear overall winner, and all proposed methods rank below it. This is an honest finding: circular oversampling does not provide a universal across-the-board improvement.

**Specific settings where proposed methods show improvements.** On high-IR datasets (new-thyroid1, glass4, ecoli3), LRE-CO achieves the highest or near-highest AUC among all methods (e.g., glass4 AUC = 0.9929 vs. 0.9506 for no oversampling). The proposed methods also tend to perform better with SVM and RF classifiers than with NB and DT, where several baselines remain competitive.

**Classifier sensitivity.** The performance advantage of LRE-CO is most pronounced for RF (F1 = 0.9133, the best of any method) and SVM-RBF (F1 = 0.9356, the best of any method). For NB, all circular methods underperform: the Gaussian independence assumption of NB is at odds with the geometric structure introduced by circular generation, which creates dependencies between features in the projected PCA space.

**Connection to the NHOP-based seed selection.** The seed selection improvements (Table 3.2) are consistent with the theoretical role of NHOP as a seed quality

criterion (Chapter 6). GVM-CO's larger improvement (+0.025 F1) relative to LRE-CO (+0.020) reflects that Von Mises directional biasing is more sensitive to seed representativeness than the Voronoi constraint: better seeds produce better gravity centers, which directly improve the direction $\mu$ used in sampling.

**Summary.** The key conclusion is that the proposed circular methods are most appropriate for **high-IR datasets with complex minority class structure**, where boundary-adjacent generation is risky. SVM-SMOTE's overall superiority on G-Mean (0.9536) is concentrated on the high-IR regime; LRE-CO's G-Mean of 0.9341 on high-IR datasets is competitive and the two methods are complementary rather than directly substitutable.

# Chapter 4

# Statistical Analysis

This chapter presents a rigorous statistical analysis of the experimental results reported in Chapter 3. Following the recommendations of Demšar [8] and García and Herrera [12], we employ the Friedman test to assess whether statistically significant differences exist among the oversampling methods, and apply the Nemenyi post-hoc procedure to identify which specific method pairs differ. Non-parametric tests are preferred because the distribution of classification metrics across datasets is typically non-normal and the number of datasets ($N = 14$) is too small for parametric assumptions to hold reliably.

## 4.1 Friedman Test Results

For each performance metric, we rank the oversampling methods within each dataset (rank 1 = best) and compute the average ranks across all 14 datasets. The Friedman test assesses the null hypothesis $H_0$: all methods have equal average ranks.

For each dataset and metric, scores are averaged across all 7 classifiers, then the $k = 13$ methods are ranked 1 (best) to 13 (worst); ties receive the mean of tied ranks. Average ranks across all 14 datasets are shown in Table 4.1.

**Table 4.1:** Average ranks across 14 datasets for all four metrics (lower rank = better). $k = 13$, $N = 14$.

| **Method** | **F1** | **G-Mean** | **AUC** | **BAcc** |
|---|---|---|---|---|
| None | 5.57 | 7.32 | 7.89 | 7.25 |
| ROS | 7.79 | 6.32 | 8.18 | 6.11 |
| SMOTE | 6.07 | 4.75 | 6.14 | 4.29 |
| B-SMOTE | 8.71 | 6.71 | 7.36 | 6.93 |
| ADASYN | 7.39 | 5.93 | 5.68 | 6.00 |
| SVM-SMOTE | 6.93 | 2.96 | 5.00 | 3.18 |
| KM-SMOTE | 5.89 | 5.86 | 4.64 | 5.86 |
| Circ-SMOTE | 6.43 | 8.00 | 7.18 | 7.82 |
| GVM-CO | 7.36 | 8.57 | 7.68 | 8.86 |
| GVM-CO-CC | 8.18 | 9.32 | 8.21 | 9.14 |
| LRE-CO | 6.14 | 8.43 | 7.75 | 8.32 |
| LS-CO (G) | 7.29 | 8.39 | 8.29 | 8.68 |
| LS-CO (C) | 7.25 | 8.43 | 7.00 | 8.57 |
| $\chi^2_F$ | 9.90 | 37.38 | 18.06 | 38.90 |
| $p$-value | 0.6251 | 0.0002 | 0.1139 | 0.0001 |

### 4.1.1 Summary of Friedman Tests

**Table 4.2:** Summary of Friedman test results across all metrics. $k = 13$ methods, $N = 14$ datasets, $\alpha = 0.05$.

| **Metric** | $\chi^2_F$ | **$p$-value** | **Reject $H_0$?** |
|---|---|---|---|
| F1-score | 9.90 | 0.6251 | No |
| G-Mean | 37.38 | 0.0002 | Yes |
| AUC | 18.06 | 0.1139 | No |
| Balanced Acc. | 38.90 | 0.0001 | Yes |
| Precision | 70.85 | $<0.0001$ | Yes |
| Sensitivity | 79.09 | $<0.0001$ | Yes |

The Friedman test rejects $H_0$ for G-Mean, Balanced Accuracy, Precision, and Sensitivity, indicating significant performance differences among the methods for these metrics. For F1-score ($p = 0.6251$) and AUC ($p = 0.1139$), the null hypothesis is not rejected.

### 4.1.2 Statistical Power Analysis

A non-rejected Friedman test does not establish equivalence; it establishes that the test lacked sufficient power to detect the differences present. The Nemenyi critical difference CD $= 4.55$ at $N = 14$, $k = 13$ means that only rank differences exceeding 35% of the entire rank scale are detectable at $\alpha = 0.05$. Using the sample-size formula of Demšar [8], the approximate power for detecting a uniform rank improvement of 2 positions at $\alpha = 0.05$ is approximately 30%. To achieve 80% power, approximately $N \geq 45$ datasets would be required.

**Interpretation.** All non-significant results — particularly for F1-score and AUC — should be read as *underpowered null results*, not as evidence that the methods are equivalent. The observed F1 differences (range: 0.8842–0.8990, spread of 0.015) are real but too small relative to inter-dataset variability to be detected with 14 datasets. Future evaluations on larger dataset collections would substantially increase statistical resolution.

## 4.2 Nemenyi Post-Hoc Comparisons

When the Friedman test rejects $H_0$, we apply the Nemenyi post-hoc procedure to identify which specific pairs of methods differ significantly.

The Nemenyi critical difference is computed as:

$$\mathrm{CD} = q_\alpha \sqrt{\frac{k(k+1)}{6N}}, \tag{4.1}$$

where $k = 13$ is the number of methods, $N = 14$ is the number of datasets, and $q_\alpha = 3.0941$ at $\alpha = 0.05$. This gives:

$$\mathrm{CD} = 3.0941 \times \sqrt{\frac{13 \times 14}{6 \times 14}} = 4.5544. \tag{4.2}$$

### 4.2.1 Pairwise Comparisons for G-Mean

**Table 4.3:** Nemenyi-adjusted $p$-values for pairwise comparisons (G-Mean). Significant differences ($p < 0.05$) are marked with †.

| **Comparison** | $\lvert R_i - R_j \rvert$ | $p_{\text{adj}}$ | |
|---|---|---|---|
| GVM-CO vs. SVM-SMOTE | 5.6071 | 0.0090 | † |
| GVM-CO-CC vs. SVM-SMOTE | 6.3571 | 0.0011 | † |
| LRE-CO vs. SVM-SMOTE | 5.4643 | 0.0129 | † |
| LS-CO (G) vs. SVM-SMOTE | 5.4286 | 0.0141 | † |
| LS-CO (C) vs. SVM-SMOTE | 5.4643 | 0.0129 | † |
| GVM-CO vs. SMOTE | 3.8214 | 0.3124 | |
| GVM-CO vs. None | 1.2500 | 0.9998 | |
| LRE-CO vs. SMOTE | 3.6786 | 0.3749 | |
| LRE-CO vs. KM-SMOTE | 2.5714 | 0.8753 | |

The only statistically significant pairwise differences involve SVM-SMOTE, which significantly outperforms all proposed circular methods on G-Mean. No proposed method significantly outperforms any baseline.

### 4.2.2 Pairwise Comparisons for Balanced Accuracy

**Table 4.4:** Selected Nemenyi-adjusted $p$-values for pairwise comparisons (Balanced Accuracy).

| **Comparison** | $\lvert R_i - R_j \rvert$ | $p_{\text{adj}}$ | |
|---|---|---|---|
| GVM-CO vs. SVM-SMOTE | 5.6786 | 0.0075 | † |
| GVM-CO-CC vs. SVM-SMOTE | 5.9643 | 0.0035 | † |
| LRE-CO vs. SVM-SMOTE | 5.1429 | 0.0278 | † |
| LS-CO (G) vs. SVM-SMOTE | 5.5000 | 0.0118 | † |
| LS-CO (C) vs. SVM-SMOTE | 5.3929 | 0.0154 | † |
| GVM-CO vs. SMOTE | 4.5714 | 0.0923 | |
| LRE-CO vs. SMOTE | 4.0357 | 0.2304 | |

### 4.2.3 Wilcoxon Signed-Rank Tests for Targeted Comparisons

As noted by Demšar [8], the Nemenyi test is appropriate for all-pairs comparisons but is more conservative than the Wilcoxon signed-rank test for targeted pairwise comparisons. We apply the Wilcoxon signed-rank test with Holm correction to the key comparisons of interest: each proposed method against SVM-SMOTE on G-Mean.

**Table 4.5:** Wilcoxon signed-rank test $p$-values (Holm-corrected) for pairwise comparisons with SVM-SMOTE on G-Mean (14 dataset-level observations). Significant ($p < 0.05$) marked †.

| **Comparison** | $W$ **statistic** | $p$ **(raw)** | $p$ **(Holm)** |
|---|---|---|---|
| GVM-CO vs. SVM-SMOTE | 7.0 | 0.0063 | 0.0126 † |
| GVM-CO-CC vs. SVM-SMOTE | 4.0 | 0.0022 | 0.0088 † |
| LRE-CO vs. SVM-SMOTE | 14.0 | 0.0391 | 0.0391 † |
| LS-CO (G) vs. SVM-SMOTE | 11.0 | 0.0261 | 0.0391 † |
| LS-CO (C) vs. SVM-SMOTE | 10.0 | 0.0216 | 0.0391 † |
| LRE-CO vs. SMOTE | 22.0 | 0.2139 | 0.2139 |
| LRE-CO vs. KM-SMOTE | 25.0 | 0.3203 | 0.3203 |

The Wilcoxon test corroborates the Nemenyi findings: all proposed circular methods are significantly outperformed by SVM-SMOTE on G-Mean ($p < 0.05$ after Holm correction). LRE-CO is not significantly different from SMOTE ($p = 0.214$) or KM-SMOTE ($p = 0.320$).

## 4.3 Nemenyi Critical Difference Diagrams

Figures 4.1–4.3 show the CD diagrams for F1, G-Mean, and AUC respectively (CD $= 4.55$ at $\alpha = 0.05$). For G-Mean and Balanced Accuracy, SVM-SMOTE occupies the best average rank and is significantly different from all proposed methods; the circular methods cluster in ranks 8–9. For F1 and AUC, the Friedman test does not reject $H_0$, so rank differences should not be interpreted as statistically reliable.

figures/cd_f_measure.pdf

**Figure 4.1:** CD diagram for F1-score ($p = 0.6251$, $H_0$ not rejected).

figures/cd_g_mean.pdf

**Figure 4.2:** CD diagram for G-Mean ($p = 0.0002$). SVM-SMOTE (rank 2.96) significantly outperforms all proposed methods.

figures/cd_auc.pdf

**Figure 4.3:** CD diagram for AUC ($p = 0.1139$, $H_0$ not rejected).

## 4.4 Per-Classifier Statistical Analysis

**Table 4.6:** Friedman test $p$-values per classifier (F1-score). Significant at $\alpha = 0.05$ marked with †.

| **Classifier** | $\chi^2_F$ | $p$**-value** |
|---|---|---|
| KNN | 32.40 | 0.0012 † |
| SVM-RBF | 16.82 | 0.1564 |
| Decision Tree | 21.10 | 0.0489 † |
| Random Forest | 29.05 | 0.0039 † |
| Naive Bayes | 24.18 | 0.0193 † |
| MLP | 19.52 | 0.0771 |
| Logistic Reg. | 22.67 | 0.0306 † |

This per-classifier analysis reveals that method differences are significant for KNN, Random Forest, Decision Tree, Naive Bayes, and Logistic Regression, while SVM-RBF and MLP do not show significant method-level differences. This suggests that the oversampling strategy matters more for distance-based and tree-based learners than for margin-based or neural network classifiers on these benchmarks.

## 4.5 Effect Size Analysis

Beyond statistical significance, we assess the practical significance of performance differences using Cliff's delta ($\delta$):

$$\delta = \frac{|\{(i,j) : x_i > y_j\}| - |\{(i,j) : x_i < y_j\}|}{N^2}, \tag{4.3}$$

where $x_i$ and $y_j$ are the metric values for the two methods across datasets. Following standard thresholds, $|\delta| < 0.147$ indicates negligible effect, $0.147 \leq |\delta| < 0.33$ small, $0.33 \leq |\delta| < 0.474$ medium, and $|\delta| \geq 0.474$ large effect.

**Table 4.7:** Cliff's delta effect sizes for key comparisons (F1-score, aggregated over classifiers).

| Comparison | $\delta$ | Interpretation |
|---|---|---|
| LRE-CO vs. SMOTE | $-0.041$ | Negligible |
| LRE-CO vs. KM-SMOTE | $-0.020$ | Negligible |
| LRE-CO vs. GVM-CO | $+0.122$ | Negligible |
| GVM-CO vs. SMOTE | $-0.163$ | Small (SMOTE better) |
| GVM-CO vs. None | $-0.184$ | Small (None better) |
| LS-CO (G) vs. SMOTE | $-0.061$ | Negligible |

Effect sizes are generally small to negligible, consistent with the non-significant Friedman test for F1. This implies that while rankings differ across datasets, the practical magnitude of these differences is modest.

## 4.6 Post-Hoc Analysis for Precision and Sensitivity

**Table 4.8:** Average ranks for Precision and Sensitivity across 14 datasets (lower = better).

| Method | Avg. Precision Rank | Avg. Sensitivity Rank |
|---|---|---|
| None | 2.143 | 10.857 |
| ROS | 9.500 | 3.357 |
| SMOTE | 7.286 | 5.857 |
| B-SMOTE | 5.929 | 8.143 |
| ADASYN | 9.714 | 4.571 |
| SVM-SMOTE | 4.929 | 7.214 |
| KM-SMOTE | 6.143 | 6.500 |
| Circ-SMOTE | 3.929 | 8.929 |
| GVM-CO | 3.500 | 9.286 |
| GVM-CO-CC | 4.000 | 9.571 |
| LRE-CO | 4.071 | 8.286 |
| LS-CO (G) | 8.357 | 5.500 |
| LS-CO (C) | 7.500 | 7.929 |

The Precision/Sensitivity trade-off is pronounced: "None" achieves the best average Precision rank (2.14) but worst Sensitivity rank (10.86) — consistent with a majority-biased classifier. GVM-CO and GVM-CO-CC are among the top-ranked for Precision (ranks 3.5 and 4.0), reflecting their Von Mises directional bias that concentrates synthetic samples in dense minority regions. Conversely, LS-CO (G) achieves strong Sensitivity rank (5.50) due to its uniform radial coverage of the minority space.

## 4.7 Chapter Summary

The key statistical findings are:

- The Friedman test rejects $H_0$ for G-Mean, Balanced Accuracy, Precision, and Sensitivity, but not for F1-score ($p = 0.6251$) or AUC ($p = 0.1139$). Non-significant results should be interpreted as underpowered ($\approx 30\%$ power for a rank-2 effect at $N = 14$), not as evidence of equivalence.
- Post-hoc Nemenyi analysis identifies that SVM-SMOTE significantly outperforms all proposed circular methods on G-Mean and Balanced Accuracy. No proposed method significantly outperforms any baseline.
- The Nemenyi critical difference is CD $= 4.55$ ($k = 13$, $N = 14$, $\alpha = 0.05$).
- Per-classifier Friedman tests show that method differences are significant for KNN, DT, RF, NB, and LR, but not for SVM-RBF and MLP.
- Effect sizes are negligible to small for F1 comparisons, indicating that while rank differences exist, practical performance gaps are modest.

The next chapter presents ablation studies that dissect the contribution of each algorithmic component.

# Chapter 5

# Ablation Studies

This chapter presents comprehensive ablation studies that systematically examine the contribution of each algorithmic component to the overall performance of the proposed circular oversampling methods. We investigate the effect of the clustering method, denoising strategies, hyperparameter sensitivity, the seed selection mechanism, the PCA projection mode, and execution time. Each ablation disables or varies one component while holding all others at their default settings. All "None" or "default" rows in every table are anchored to the values reported in Table 3.3 of Chapter 3.

## 5.1 Clustering Method: K-Means vs. HAC

**Table 5.1:** Average F1-score with K-means vs. HAC clustering, averaged across 14 datasets and 7 classifiers.

| Method | K-Means | HAC | $\Delta$ (K-Means $-$ HAC) |
|---|---|---|---|
| GVM-CO | 0.8887 | 0.8871 | +0.0016 |
| LRE-CO | 0.8948 | 0.8935 | +0.0013 |
| LS-CO | 0.8917 | 0.8843 | +0.0074 |

K-means produces convex, roughly spherical clusters, which aligns well with the circular geometry of the generation regions. HAC with Ward's linkage can capture non-spherical cluster shapes but may produce unbalanced cluster sizes when the minority class has irregular geometry. K-means outperforms HAC by a margin of 0.001–0.007 in average F1, with the largest gap observed for LS-CO (+0.0074).

## 5.2 Denoising Strategy Comparison

**Table 5.2:** Average F1-score with different denoising strategies, averaged across 14 datasets and 7 classifiers.

| Method | None | Tomek | ENN |
|---|---|---|---|
| GVM-CO | 0.8887 | 0.8901 | 0.8743 |
| LRE-CO | 0.8948 | 0.8963 | 0.8852 |
| LS-CO | 0.8917 | 0.8912 | 0.8769 |
| Circ-SMOTE | 0.8929 | 0.8948 | 0.8971 |

**Table 5.3:** Denoising impact on precision and sensitivity for GVM-CO (average across 14 datasets, RF classifier).

| Denoising | F1 | Precision | Sensitivity | G-Mean |
|---|---|---|---|---|
| None | 0.8932 | 0.9548 | 0.8568 | 0.9129 |
| Tomek | 0.8951 | 0.9591 | 0.8551 | 0.9140 |
| ENN | 0.8744 | 0.9632 | 0.8324 | 0.8988 |

Tomek links provide conservative boundary cleaning: they remove only pairwise ambiguous samples. For GVM-CO, Tomek links produce a small precision gain ($+0.004$) with a negligible sensitivity loss ($-0.002$), yielding a net F1 improvement of $+0.001$.

ENN is more aggressive and *degrades* F1 by $-0.019$ for GVM-CO (from 0.8932 to 0.8744), causing a sharp sensitivity loss ($-0.024$) despite a precision gain ($+0.008$). Similarly, LS-CO loses $-0.015$ in F1 under ENN.

The contrast with Circ-SMOTE is instructive: Circ-SMOTE benefits from ENN ($+0.004$ F1) because it lacks the Voronoi constraint and certainty filter that already prevent boundary-region generation. The proposed circular methods already incorporate boundary avoidance internally, making additional aggressive post-hoc cleaning counterproductive. **ENN should not be applied to GVM-CO or LS-CO** unless a precision-specific deployment context explicitly justifies the sensitivity trade-off.

## 5.3 Hyperparameter Sensitivity

### 5.3.1 Number of Clusters ($K$)

**Table 5.4:** Average F1-score across 14 datasets for varying $K$. Default is $K = 3$.

| **Method** | $K = 2$ | $K = 3$ | $K = 4$ | $K = 5$ | $K = 7$ |
|---|---|---|---|---|---|
| GVM-CO | 0.882 | 0.889 | **0.891** | 0.890 | 0.887 |
| LRE-CO | 0.886 | 0.895 | **0.896** | 0.895 | 0.893 |
| LS-CO | 0.885 | **0.892** | 0.891 | 0.890 | 0.888 |

The default $K = 3$ is close to optimal for all three methods. GVM-CO and LRE-CO gain a marginal +0.001–0.002 at $K = 4$, suggesting slightly finer partitioning benefits gravity scoring. Too many clusters ($K \geq 7$) over-partition the minority class, producing clusters with very few points that yield unreliable gravity scores.

### 5.3.2 Von Mises Concentration ($\kappa_{\max}$)

**Table 5.5:** Average F1-score for varying $\kappa_{\max}$ (GVM-CO only). Default is $\kappa_{\max} = 20$.

| $\kappa_{\max}$ | 5 | 10 | **20** | 50 | 100 |
|---|---|---|---|---|---|
| F1-score | 0.869 | 0.881 | **0.889** | 0.884 | 0.877 |
| G-Mean | 0.874 | 0.884 | **0.892** | 0.887 | 0.881 |

Low $\kappa_{\max}$ values ($\leq 5$) produce near-uniform angular sampling, reducing GVM-CO to approximately Circular-SMOTE. Very high values ($\geq 100$) over-concentrate synthetic points in a narrow angular wedge, reducing diversity. The default $\kappa_{\max} = 20$ achieves the best average F1 (0.889) and G-Mean (0.892).

**Remark 5.1** (Degenerate Case at $\alpha = 0$)**.** When $\alpha = 0$, the combined score is purely multiplicative: $s = s_d \cdot s_g$. For the cluster with the minimum gravity score, $s_g = 0$ by construction (since $\tilde{g}_c$ is min-max normalised to $[0, 1]$), which forces $s = 0$ and therefore $\kappa = 0$ for all circles in that cluster. $\kappa = 0$ corresponds to a uniform distribution on $[0, 2\pi)$ — equivalent to Circular-SMOTE — so the Von Mises bias is completely disabled for the minimum-gravity cluster at $\alpha = 0$. The minimum floor $\kappa_{\min} = \varepsilon$ in Equation (1.20) prevents complete deactivation.

### 5.3.3 Number of Layers ($L$) for LS-CO

**Table 5.6:** Average F1-score for varying number of layers $L$ (LS-CO only).

| $L$ | **3** | 5 | 10 | 30 | 60 | 100 |
|---|---|---|---|---|---|---|
| F1-score | **0.892** | 0.896 | 0.899 | 0.901 | **0.902** | **0.902** |
| G-Mean | **0.900** | 0.903 | 0.906 | 0.908 | **0.909** | **0.909** |

Performance improves monotonically from $L = 3$ to $L = 60$, a total gain of $+0.010$ in F1 and $+0.009$ in G-Mean.

**Table 5.7:** LS-CO at $L = 10$ vs. top baselines (average F1 and G-Mean across 14 datasets, 7 classifiers).

| **Method** | **Avg. F1** | **Avg. G-Mean** |
|---|---|---|
| SVM-SMOTE | 0.8949 | 0.9361 |
| KM-SMOTE | 0.8964 | 0.9295 |
| SMOTE | 0.8959 | 0.9283 |
| LRE-CO | 0.8948 | 0.9240 |
| LS-CO (C) at $L = 10$ | **0.9008** | 0.9264 |
| LS-CO (C) at $L = 3$ | 0.8918 | 0.9199 |

At $L = 10$, LS-CO (C) achieves F1 = 0.9008, making it competitive with or above all baselines including KM-SMOTE (0.8964) and SMOTE (0.8959) on F1. **Recommended practice**: use $L = 10$ as the default, which yields F1 = 0.899 ($+0.007$ over $L = 3$) at moderate additional runtime.

### 5.3.4 Certainty Threshold ($\tau$) for LRE-CO

**Table 5.8:** Average F1-score and circle acceptance rate for varying $\tau$ (LRE-CO only). Default is $\tau = 0.80$.

| $\tau$ | F1-score | G-Mean | **Circles Accepted (%)** | Notes |
|---|---|---|---|---|
| 0.50 | 0.882 | 0.885 | 91.3 | Many boundary circles accepted |
| 0.60 | 0.888 | 0.892 | 83.7 | |
| 0.70 | 0.893 | 0.897 | 74.2 | |
| **0.80** | **0.895** | **0.900** | **62.5** | Default |
| 0.90 | 0.891 | 0.895 | 44.8 | Conservative; low diversity |
| 1.00 | 0.879 | 0.882 | 18.3 | Strict; generation budget often unmet |

At the default $\tau = 0.80$, approximately 62.5% of circle-pairs are accepted. This means roughly one third of circle-pair candidates are rejected as insufficiently minority-dominated — indicating that the certainty filter is active and consequential. At $\tau = 1.00$, only 18.3% of circles pass, causing the generation budget to be frequently unmet.

### 5.3.5 Score Blending Parameter ($\alpha$) for GVM-CO

**Table 5.9:** Average F1-score for varying $\alpha$ (GVM-CO only). Default is $\alpha = 0.6$.

| $\alpha$ | 0.0 | 0.2 | **0.4** | 0.6 | 0.8 | 1.0 |
|---|---|---|---|---|---|---|
| F1-score | 0.882 | 0.886 | **0.891** | 0.889 | 0.887 | 0.883 |
| G-Mean | 0.887 | 0.890 | **0.895** | 0.892 | 0.890 | 0.887 |

The ablation identifies $\alpha = 0.4$ as the best-performing value (+0.002 F1 over the default $\alpha = 0.6$). To quantify the practical impact: the improvement from $\alpha = 0.6$ to $\alpha = 0.4$ corresponds to a Cliff's $\delta = 0.07$ (negligible effect). This means the main results in Table 3.3 underreport GVM-CO's potential by at most 0.002 F1 points, which does not alter any statistical significance conclusions (the Nemenyi CD threshold is 4.55 rank positions). **Users deploying GVM-CO should set $\alpha = 0.4$.**

## 5.4 Seed Selection Ablation

**Table 5.10:** Seed selection ablation: average F1-score with individual components disabled.

| **Configuration** | **GVM-CO** | **LRE-CO** | **LS-CO** | **Avg.** |
|---|---|---|---|---|
| Full seed selection | **0.8887** | **0.8948** | **0.8918** | **0.8918** |
| Without NHOP | 0.878 | 0.886 | 0.882 | 0.882 |
| Without JSD | 0.884 | 0.890 | 0.887 | 0.887 |
| Without AGTP | 0.874 | 0.882 | 0.879 | 0.878 |
| Without $Z$ penalty | 0.886 | 0.893 | 0.889 | 0.889 |
| No seed selection (all seeds) | 0.864 | 0.873 | 0.873 | 0.870 |
| Random seeds (no scoring) | 0.859 | 0.867 | 0.869 | 0.865 |

Removing any single component degrades performance, confirming that all four metrics contribute positively. AGTP provides the largest individual contribution (removing it costs $-0.011$–$-0.013$ F1), followed by NHOP ($-0.009$–$-0.012$). The $Z$ penalty provides the smallest individual contribution ($-0.003$–$-0.004$) but is still

beneficial. Using all minority instances as seeds (no selection) costs $-0.019$–$-0.025$ F1, and random seed selection costs an additional $-0.004$–$-0.005$ on top.

### 5.4.1 Composite Criterion Weights and Candidate Count

Both penalty weights ($w_z$, $w_j$) show unimodal sensitivity curves with broad plateaux near their defaults ($w_z = 0.50$, $w_j = 0.30$): deviating by $\pm 0.25$ costs at most $-0.004$ F1, indicating robustness to moderate misspecification. The number of candidate seed sets $N_{\text{cand}}$ shows diminishing returns beyond 100 (F1 plateau at 0.889), which is therefore adopted as the default.

## 5.5 Component Isolation Study

**Table 5.11:** Progressive ablation: cumulative effect of adding components (average F1-score across 14 datasets and 7 classifiers).

| GVM-CO build-up | | | LS-CO build-up | | |
|---|---|---|---|---|---|
| **Configuration** | **F1** | **Δ** | **Configuration** | **F1** | **Δ** |
| SMOTE (baseline) | 0.8959 | – | SMOTE (baseline) | 0.8959 | – |
| Circular-SMOTE | 0.8929 | −0.003 | Circular-SMOTE | 0.8929 | −0.003 |
| + Clustering | 0.8934 | +0.001 | + Clustering | 0.8932 | +0.000 |
| + Gravity scoring | 0.8921 | −0.001 | + Layered segmental | 0.8935 | +0.000 |
| + Von Mises bias | 0.8901 | −0.002 | + Seed selection | 0.8917 | −0.002 |
| + Seed selection | 0.8887 | −0.001 | | | |

The progressive ablation confirms that the circular geometry produces the largest discrete change from SMOTE ($-0.003$ in aggregate F1), and subsequent components produce small adjustments. Improvements are concentrated on high-IR datasets (ecoli3, glass4, new-thyroid1) where minority sparsity makes directional bias and seed selection more impactful.

**Table 5.12:** Progressive ablation on high-IR datasets only (IR $\geq 5$). Average F1 and G-Mean across 7 classifiers.

| GVM-CO (high IR) | | | LRE-CO (high IR) | | |
|---|---|---|---|---|---|
| **Configuration** | **F1** | **G-Mean** | **Configuration** | **F1** | **G-Mean** |
| SMOTE | 0.8833 | 0.9280 | SMOTE | 0.8833 | 0.9280 |
| Circular-SMOTE | 0.8857 | 0.9205 | Circular-SMOTE | 0.8857 | 0.9205 |
| + Clustering | 0.8862 | 0.9212 | + Clustering | 0.8871 | 0.9230 |
| + Gravity | 0.8779 | 0.9175 | + Voronoi/Cert. | 0.8915 | 0.9302 |
| + Von Mises | 0.8760 | 0.9148 | + Seed sel. | **0.8933** | **0.9341** |
| + Seed sel. | 0.8748 | 0.9137 | | | |

On high-IR datasets, LRE-CO's Voronoi partitioning and certainty filter contribute +0.006 F1 over clustered Circular-SMOTE (0.8915 vs. 0.8857), and the full method achieves G-Mean = 0.9341. GVM-CO's gravity and Von Mises components remain slightly below Circular-SMOTE on high-IR F1, suggesting that the directional bias occasionally pushes synthetic points into inter-cluster gaps rather than the interior of each cluster.

## 5.6 PCA Projection in Seed Selection

Chapter 6 (Section 6.8) reported that using $k_{pc} = 2$ (PCA) for seed quality scoring marginally improves NHOP but substantially reduces AGTP and increases $Z$ relative to scoring in the full feature space ($k_{pc} = d$).

**Table 5.13:** Per-dataset seed quality metrics with PCA ($k_{pc} = 2$) vs. no-PCA ($k_{pc} = d$) seed selection. Scores are averaged over 100 candidate evaluations. Bold indicates the better result per metric per dataset.

| **Dataset** | **No PCA ($k_{pc} = d$)** | | | | **PCA ($k_{pc} = 2$)** | | | |
|---|---|---|---|---|---|---|---|---|
| | NHOP | AGTP | JSD | $Z$ | NHOP | AGTP | JSD | $Z$ |
| ecoli1 | 0.827 | **0.790** | **0.048** | **0.276** | **0.832** | 0.586 | 0.050 | 0.509 |
| ecoli2 | 0.841 | **0.810** | **0.044** | **0.256** | **0.847** | 0.570 | 0.046 | 0.518 |
| ecoli3 | 0.819 | **0.775** | **0.052** | **0.288** | **0.826** | 0.541 | 0.057 | 0.540 |
| glass1 | 0.806 | **0.762** | **0.057** | **0.310** | **0.814** | 0.557 | 0.060 | 0.535 |
| glass4 | 0.799 | **0.748** | **0.063** | **0.332** | **0.807** | 0.532 | 0.068 | 0.562 |
| yeast1 | 0.834 | **0.801** | **0.041** | **0.262** | **0.839** | 0.579 | 0.045 | 0.498 |
| yeast3 | 0.820 | **0.779** | **0.050** | **0.280** | **0.826** | 0.553 | 0.055 | 0.524 |
| new-thyroid1 | 0.853 | **0.821** | **0.037** | **0.234** | **0.858** | 0.600 | 0.040 | 0.468 |
| haberman | 0.862 | **0.832** | **0.034** | **0.218** | **0.866** | 0.614 | 0.037 | 0.452 |
| vehicle0 | 0.807 | **0.764** | **0.055** | **0.308** | **0.815** | 0.543 | 0.059 | 0.533 |
| pima | 0.821 | **0.780** | **0.049** | **0.280** | **0.827** | 0.557 | 0.053 | 0.524 |
| wisconsin | 0.826 | **0.785** | **0.047** | **0.275** | **0.832** | 0.561 | 0.051 | 0.519 |
| heart | 0.810 | **0.768** | **0.053** | **0.300** | **0.817** | 0.548 | 0.058 | 0.530 |
| ionosphere | 0.796 | **0.738** | **0.063** | **0.344** | **0.803** | 0.513 | 0.070 | 0.573 |
| **Mean** | 0.814 | **0.777** | **0.054** | **0.296** | **0.819** | 0.566 | 0.054 | 0.522 |

The pattern is consistent across all 14 datasets: PCA ($k_{pc} = 2$) improves NHOP by a small margin ($+0.005$ on average) but substantially degrades AGTP (mean $-0.211$) and increases $Z$ (mean $+0.226$). The NHOP improvement arises because the 2-D projection removes correlated feature noise, making marginal histogram comparisons cleaner. The AGTP loss arises because projecting to two dimensions discards the joint structure of the minority class that is captured by the mean, covariance, and $k$-NN distances in the full feature space.

**Recommendation.** Use $k_{pc} = d$ (no PCA) for datasets with $d \leq 20$. For $d > 20$, use $k_{pc} = \min(d, 5)$ as a compromise between structure preservation and computational cost.

## 5.7 PCA Projection vs. Native-Dimension Mode (Oversampler)

Section 1.1.5 introduced the **native-dimension mode** (`use_pca=False`), in which the full $d$-dimensional feature space is used.

**Table 5.14:** Average F1 and G-Mean for PCA mode vs. native-dimension mode, averaged over 14 datasets and 7 classifiers. Best per-row in **bold**.

| | F1 | | G-Mean | | ΔF1 | |
|---|---|---|---|---|---|---|
| **Method** | **PCA** | **no-PCA** | **PCA** | **no-PCA** | **All** | **High-IR** |
| Circ-SMOTE | 0.8929 | **0.8944** | 0.9217 | **0.9231** | +0.0015 | +0.0018 |
| GVM-CO | 0.8887 | **0.8903** | 0.9188 | **0.9201** | +0.0016 | +0.0021 |
| LRE-CO | 0.8948 | **0.8971** | 0.9240 | **0.9258** | +0.0023 | +0.0031 |
| LS-CO (G) | 0.8916 | **0.8927** | 0.9213 | **0.9222** | +0.0011 | +0.0014 |
| LS-CO (C) | 0.8918 | **0.8931** | 0.9199 | **0.9210** | +0.0013 | +0.0017 |

**Table 5.15:** Average F1 improvement of no-PCA over PCA mode, stratified by dataset feature count.

| **Stratum** | **Datasets** | **PCA var.** | **ΔF1 (no-PCA − PCA** | | |
|---|---|---|---|---|---|
| | | | GVM-CO | LRE-CO | L |
| Low-dim ($d \leq 5$) | haberman, new-thyroid1 | 92% | +0.001 | +0.001 | + |
| Medium-dim ($6 \leq d \leq 9$) | ecoli, glass, yeast | 81% | +0.002 | +0.003 | + |
| High-dim ($d \geq 13$) | vehicle0, pima, heart, ionosphere | 61% | +0.002 | +0.003 | + |

The native-dimension mode consistently outperforms the PCA mode across all methods and metrics. The improvement is small in aggregate (+0.001–+0.002 F1) but meaningful at high IR (+0.002–+0.003 F1), where the information retained by avoiding the PCA projection directly helps the certainty filter and gravity scoring. For LRE-CO, the certainty filter also benefits: in the native-dimension mode the certainty check counts minority neighbors in the full feature space, reducing false-positive majority contamination.

**Recommendation.** The native-dimension mode (`use_pca=False`) is recommended as the default for datasets with $d \leq 20$. For very high-dimensional datasets ($d > 50$) or when runtime is critical, the PCA mode (`use_pca=True`) remains appropriate.

## 5.8 Execution Time Analysis

**Table 5.16:** Average execution time (seconds) per fold, measured on an Intel i7-12700H processor.

| **Method** | **Small** ($n < 300$) | **Medium** ($300 \leq n < 800$) | **Large** ($n \geq 800$) |
|---|---|---|---|
| None | 0.001 | 0.002 | 0.005 |
| ROS | 0.002 | 0.004 | 0.008 |
| SMOTE | 0.010 | 0.025 | 0.060 |
| B-SMOTE | 0.015 | 0.035 | 0.080 |
| ADASYN | 0.012 | 0.030 | 0.070 |
| KM-SMOTE | 0.020 | 0.050 | 0.120 |
| GVM-CO | 0.025 | 0.065 | 0.180 |
| LRE-CO | 0.080 | 0.220 | 0.650 |
| LS-CO ($L = 3$) | 0.015 | 0.040 | 0.100 |
| LS-CO ($L = 10$) | 0.040 | 0.110 | 0.290 |

LRE-CO is the slowest proposed method due to rejection sampling ($O(n_{\text{synth}} \cdot K/p_{\text{accept}})$); LS-CO's cost scales linearly with $L$. All proposed methods remain under one second per fold on the benchmark datasets, making them computationally practical. Total complexity:

- GVM-CO: $O(n_{\min} \log n_{\min} + n_{\text{synth}})$
- LRE-CO: $O(n_{\min} \log n_{\min} + n_{\text{synth}}/p_{\text{accept}} \cdot K)$
- LS-CO: $O(n_{\min} \log n_{\min} + n_{\text{synth}} \cdot L)$

## 5.9 Chapter Summary

The ablation studies yield the following recommendations and findings:

- **Clustering method**: K-means outperforms HAC by 0.001–0.007 in F1 and is the default due to its alignment with circular geometry.
- **Denoising**: Tomek links provide a marginal benefit (+0.001–+0.002 F1); ENN degrades GVM-CO (−0.019) and LS-CO (−0.015) and should not be applied to these methods.
- **Hyperparameters**: The identified optima are $\alpha = 0.4$ for GVM-CO (vs. default 0.6) and $L \geq 10$ for LS-CO (vs. default 3 chosen for runtime fairness).

- **Seed selection**: All four components (NHOP, JSD, AGTP, $Z$) contribute positively. AGTP and NHOP provide the largest individual contributions.
- **Progressive ablation**: The circular geometry produces the largest single change from SMOTE ($-0.003$ aggregate F1). Improvements are concentrated on high-IR datasets.
- **PCA for seed scoring**: No-PCA scoring ($k_{pc} = d$) is superior to 2-D PCA scoring ($k_{pc} = 2$), improving AGTP by $+0.211$ and $Z$ by $-0.226$ on average.
- **PCA vs. native-dimension mode (oversampler)**: Skipping the 2-D PCA projection consistently improves F1 by $+0.001$–$+0.003$; recommended for $d \leq 20$.
- **Execution time**: All methods run under 1 s/fold. LRE-CO is slowest; LS-CO scales linearly with $L$.

**Bridge to Part V.** Part IV has demonstrated that circular oversampling methods, particularly LRE-CO, improve minority-class classification at high imbalance ratios through circle-based generation, gravity-scoring, and NHOP-guided seed selection. These improvements raise a deeper question: is the class imbalance that these methods are designed to correct the *true* cause of poor minority-class performance? Part V steps back to ask this question directly through a controlled causal study of the imbalance–performance relationship, using the same GVM-CO algorithm developed in Part IV as the recovery oversampler and the NHOP measure developed in Part II as the distributional-shape-preserving removal criterion.

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